AI-BASED HYPER-LOCAL AVIATION WEATHER FORECAST VERIFICATION SYSTEM FOR SRI LANKAN MONSOON PATTERNS



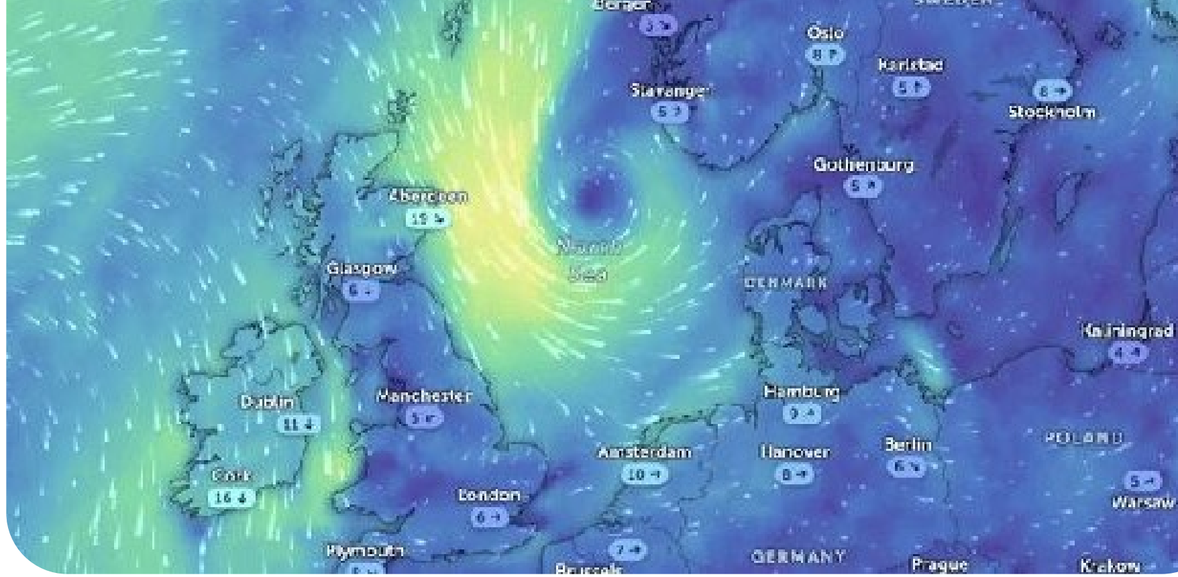
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AI-BASED HYPER-LOCAL AVIATION WEATHER FORECAST VERIFICATION SYSTEM FOR SRI LANKAN MONSOON PATTERNS

BACKGROUND

Weather forecasting plays a critical role in aviation safety and operational decision-making. Pilots depend on aviation weather reports such as TAF and METAR to obtain atmospheric conditions before take-off and landing



PROBLEM

Most aviation weather forecasting systems rely on global models such as GFS and ECMWF, which provide large-scale forecasts but It fail on Localize weather variations.

30°C
MAX. 31°C - MIN.15°C
MONDAY, MARCH 16
20% 15 KM/H WEST 1023.0

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USERS & STAKEHOLDERS

PRIMARY USERS

- Pilots
- Air Traffic Controllers
- Aviation Meteorological Officers
- Airport Operations Managers



STAKEHOLDERS

- Airport authorities (Bandaranaike International Airport)
- Civil Aviation Authority of Sri Lanka
- Airline operation management teams

30°C
MAX. 31°C - MIN.15°C
MONDAY, MARCH 16
20% 15 KM/H WEST 1023.0

USERS AND STAKEHOLDERS

How they interact with the system

- Pilots receive visibility and weather alerts before takeoff and landing
- Meteorologists verify forecasts using the AI system
- Airport authorities use the system for operational planning



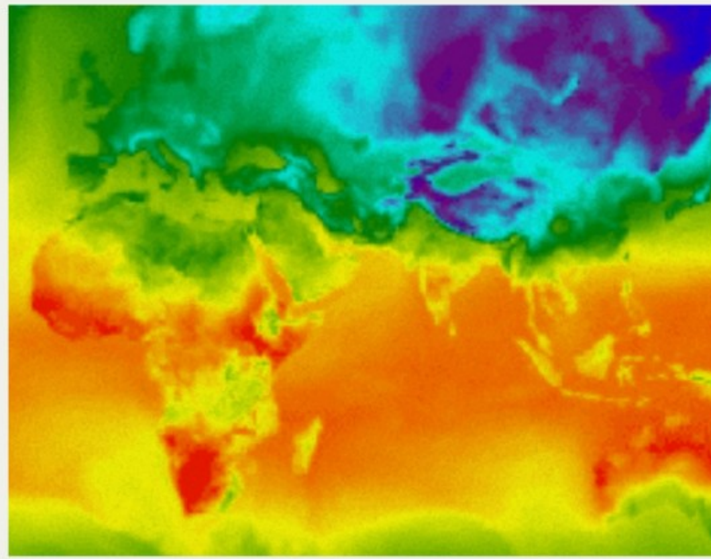
USERS AND STAKEHOLDERS

Impact of the Problem to Users and Society

- Sudden weather changes can reduce pilot visibility.
- Low cloud ceilings can affect safe aircraft landing.
- Can cause flight delays and flight diversions.
- Affects passenger travel and airline operations.



EXISTING SOLUTIONS AND THEIR LIMITATIONS



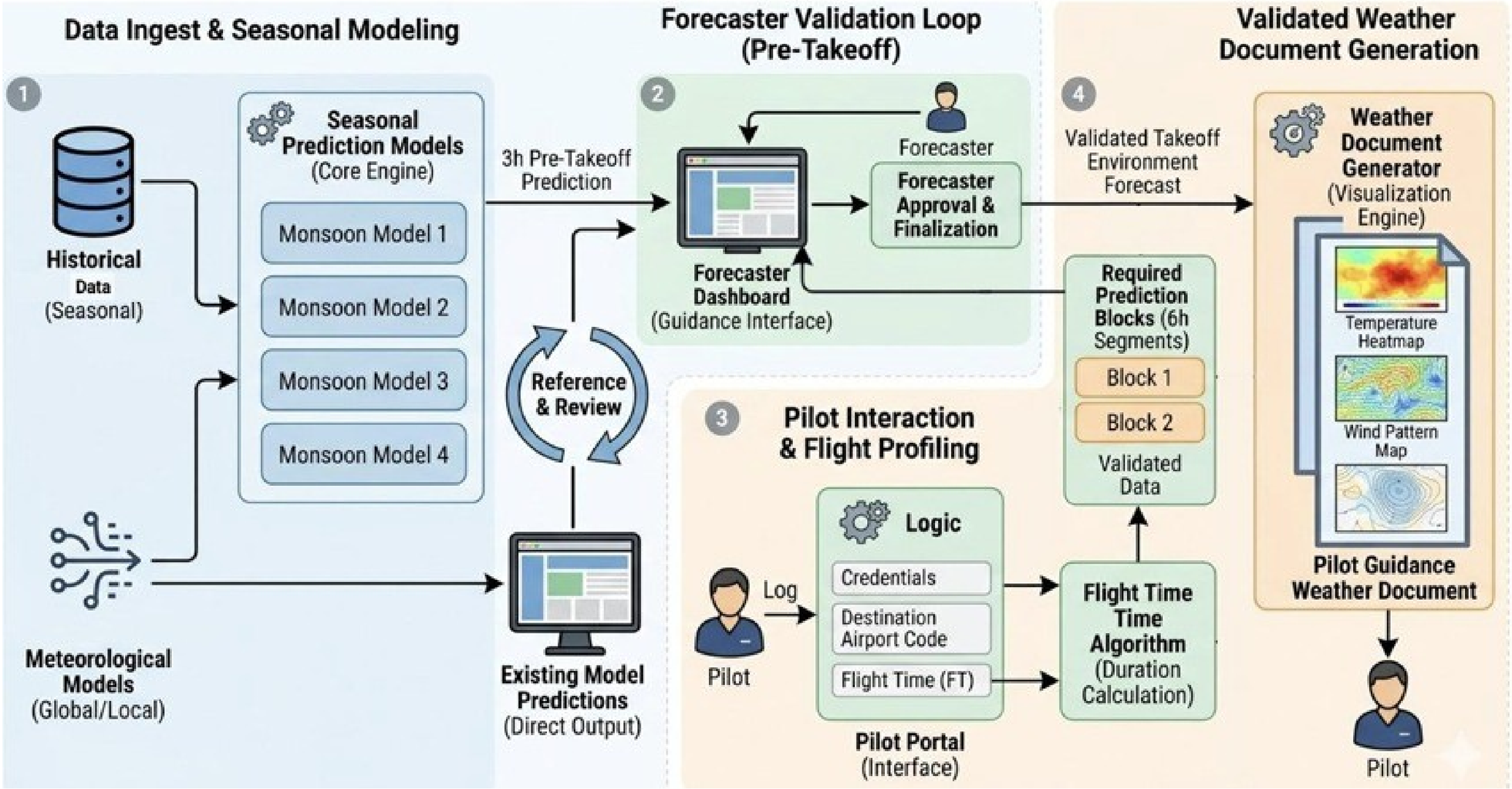
Existing Solutions

- Global Forecast System (GFS)
- ECMWF weather prediction models
- TAF and METAR reports

Limitations

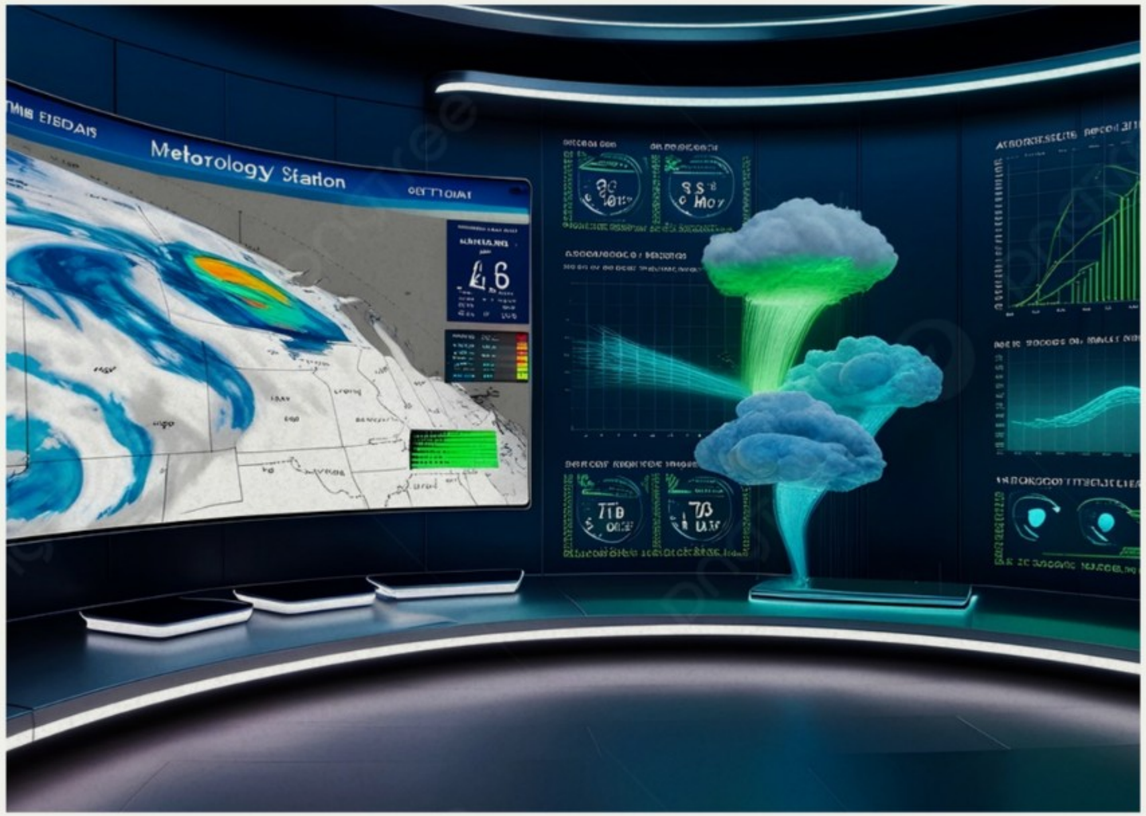
- Difficult to detect rapid local weather changes.
- Limited accuracy for airport-lev predictions.
- Cant find Past data

SYSTEM DIAGRAM



THE PROPOSED SOLUTION

- **Hyper-local AI-based aviation weather prediction system for monsoon pattons**
- **Uses machine learning to**
  analyze weather data.
- **Detects rapid weather changes near airports.**
- **Generates alerts for dangerous**



TEAM MEMBER CONTRIBUTIONS

Student	Component
Sankadeepa W.D.I	Pressure-Humidity-Temperature Prediction Model
Ranasingha R.A.A.D	Wind Pattern Prediction Model
Jayawardhana H.M.S.I	Cloud and Visibility Prediction Model for aviation Safety
Jayathilaka P.K.V.K	Relative Humidity & QNH Stability Model

SANKADEEPA W.D.I
IT22113986



Pressure-Humidity-Temperature Coupled Prediction Model

NORTHEAST MONSOON

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PRESSURE-HUMIDITY-TEMPERATURE COUPLED PREDICTION MODEL

Contribution to Overall Solution

- predict on North east monsoon
- temperature, pressure, and humidity interact with each other at airport level
- Support hyper-local aviation weather forecasting
- Develop a hybrid weather prediction model (Multilinear regression- parameter relationship & LSTM- weather pattern)

Novelty

- **Pressure-Humidity-Temperature Coupled Prediction Model**

PRESSURE-HUMIDITY-TEMPERATURE COUPLED PREDICTION MODE

Objectiv e

Develop a hyper localized model that predicts temperature, humidity and pressure together.

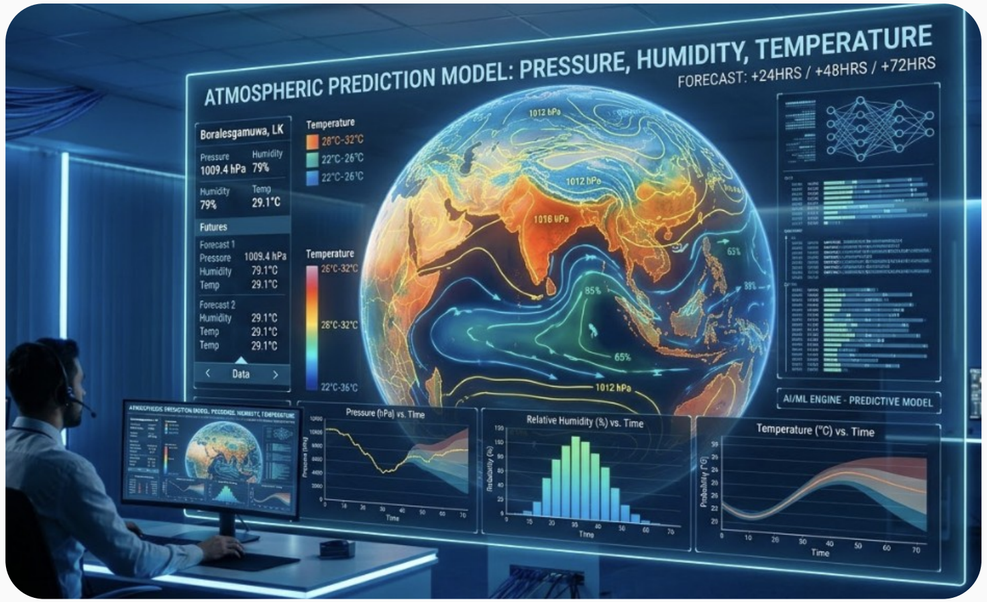
Aviation safety and decision Making

Proble m

- Northeast Monsoon causes sudden weather changes.
- Large Global model falling(gef, ecmwf)
- Fail to capture Localized data.
- Grid cell represent about 20KM and applied same prediction in all areas .

Sub-Objectives

- Collect airport weather data
- Train a multi-variable ML model
- Perform forecast verification
- Visualize validated weather patterns
  (heatmap , wind patterns)



PRESSURE-HUMIDITY-TEMPERATURE COUPLED PREDICTION MODEL

Methodology

- Aviation meteorology analysis
- Time-series machine learning
- Multi-linear regression

Technologies

- Multiple Linear Regression
- LSTM (best for weather time series)
- Neural Network

Validation and Evaluation

- Compare predictions with METAR aviation weather reports
- Test the model using historical weather data
- Compare predicted results with actual weather data
- Compare results with METAR and TAF forecasts

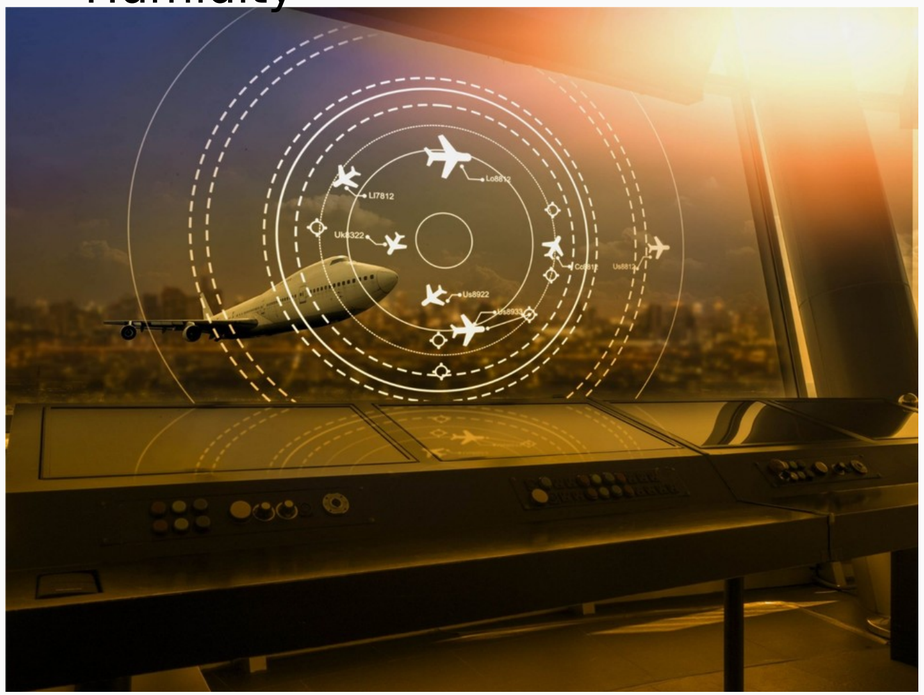
Data Availability & Ethics

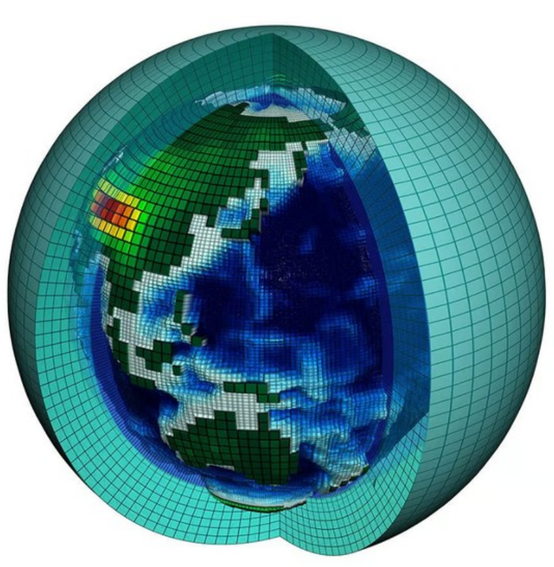
- Historical meteorological data will be used.
- Dataset includes temperature, humidity, and atmospheric pressure.
- Data obtained from meteorological or airport weather observation sources.
- No personal or sensitive information is included.

PRESSURE-HUMIDITY-TEMPERATURE COUPLED PREDICTION MODEL

System Overview

- Input → Weather Data
- Process → ML Prediction Model
- Output → Forecast Temperature, Pressure, Humidity





Dependencies

- **Airport weather datasets**
- **Forecast verification module**
- **Visualization component**

Ranasinghe R.A.D.
IT22269720



Wind Pattern Prediction Model for Southwest Monsoon

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WIND PATTERN PREDICTION MODEL FOR SOUTHWEST MONSOON

Contribution to Overall Solution

- Predict wind speed and wind direction
- Identify wind variations affecting aviation operations
- Support runway wind analysis

Novelty

- **Convert wind predictions into crosswind and headwind components**
- **Provides runway-specific aviation guidance**

WIND PATTERN PREDICTION MODEL FOR SOUTHWEST MONSOON

Problem Definition

- Wind is a critical factor in aircraft take-off and landing
- Sudden wind changes create aviation hazards
- Existing forecasts models low accuracy to predict specific wind prediction



Objective

Develop a machine learning model for wind pattern prediction during the Southwest Monsoon.

Sub-Objectives

- Analyze historical wind patterns
- Train ML models for wind prediction
- Evaluate wind forecast accuracy
- Develop pilot interaction platfome

WIND PATTERN PREDICTION MODEL FOR SOUTHWEST MONSOON

Methodology

- Time-series wind pattern analysis
- Machine learning based forecasting
- Aviation wind hazard assessment

Validation

- Compare with official aviation weather forecasts

Technologies

- Python
- Machine Learning models
- Data visualization tools

Data availability & Ethics

- Historical wind datasets
- Meteorological data only
- No personal data usage

WIND PATTERN PREDICTION MODEL FOR SOUTHWEST MONSOON

System Overview

Input → Wind data

Process → ML Wind Model

Output → Runway wind prediction



Dependencies

- Pilot portal module
- Weather dataset
- Prediction engine

Jayawardhana H.M.S.I

IT22619976



Cloud and Visibility Prediction Model for aviation Safety

First Inter-Monsoon period.

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Cloud and Visibility Prediction Model for aviation Safety

Contribution to Overall Solution

- Detect sudden weather changes that can affect aviation operations
- Monitor cloud ceiling and visibility reduction
- Improve aviation weather safety

Novelty

- **Specialized model for First Inter-Monsoon**
- **Detect fog and low cloud conditions which are the main reasons for visibility reduction in aviation**

Cloud and Visibility Prediction Model for aviation Safety

Problem Definition

- Rapid weather changes reduce pilot visibility
- Fog and low clouds can disrupt flight operation



Objective

Develop a ML model to detect sudden atmospheric changes.

Sub-Objectives

- Analyze visibility patterns using historical weather data
- Detect fog formation & sudden visibility reduction
- Train ML detection models
- Process flight time inputs that can analyze weather conditions during the

Cloud and Visibility Prediction Model for aviation Safety

Methodology

- Cloud ceiling analysis to monitor how cloud height changes over time
- Visibility pattern detection
- Machine learning classification

Validation

- Compare predictions with actual weather observations

Technologies

- Python
- Machine Learning models
- Data preprocessing tools

Data availability & Ethics

- Aviation weather dataset
- No personal data

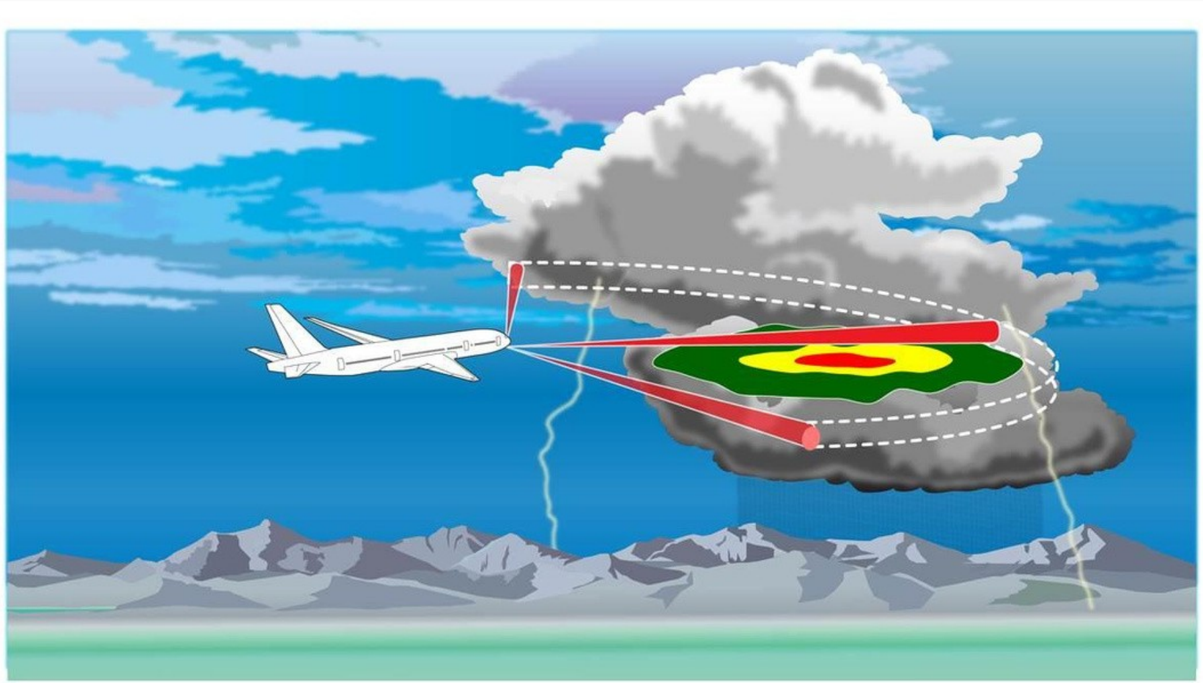
Cloud and Visibility Prediction Model for aviation Safety

System Overview

Input → Cloud & visibility data

Process → ML detection model

Output → Weather hazard alerts



Dependencies

- Flight time processing module
- Weather data

Jayathilaka P.K.V.K

IT22069504



Relative Humidity & QNH Stability Prediction Model

Second Inter-Monsoon

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RELATIVE HUMIDITY & QNH STABILITY PREDICTION MODEL

Contribution to Overall Solution

- Develop a machine learning model that Predict atmospheric stability
- Analyze relative humidity and QNH variations
- Generate pilot weather guidance documents

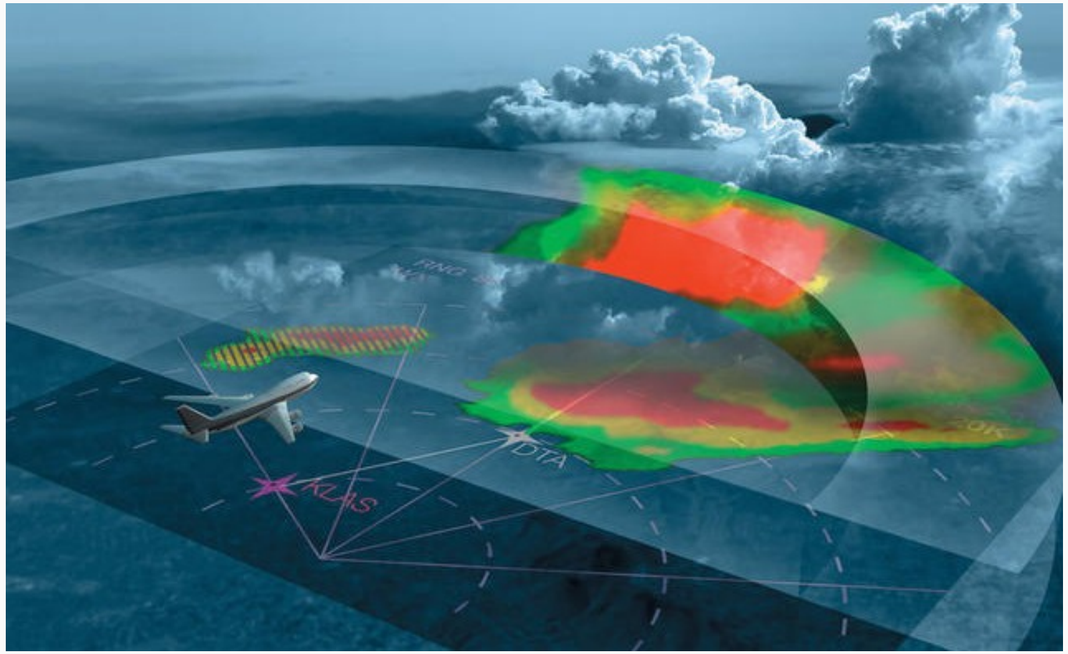
Novelty

- **Develop RH & QNH stability model**
- **Generate reliability scoring based on pressure trends**

RELATIVE HUMIDITY & QNH STABILITY PREDICTION MODEL

Problem Definition

- High humidity and QNH changes influence local weather instability
- Aviation systems lack localized pressure trend prediction



Objective

Develop a predictive model to analyze humidity peak and pressure drops.

Sub-Objectives

- Analyze humidity and QNH correlations using historical weather data
- Train ML models to predict rh & QNH variations
- Generate structured pilot weather guidance reports

RELATIVE HUMIDITY & QNH STABILITY PREDICTION MODEL

Methodology

- Time-series atmospheric analysis
- Machine learning prediction
- Data correlation modeling

Validation

- Compare predictions with METAR and TAF aviation reports

Technologies

- Python
- TensorFlow / Scikit-Learn
- Pandas / NumPy

Data availability & Ethics

- Historical weather datasets
- Meteorological data only
- Ethical compliance ensured

RELATIVE HUMIDITY & QNH STABILITY PREDICTION MODEL

System Overview

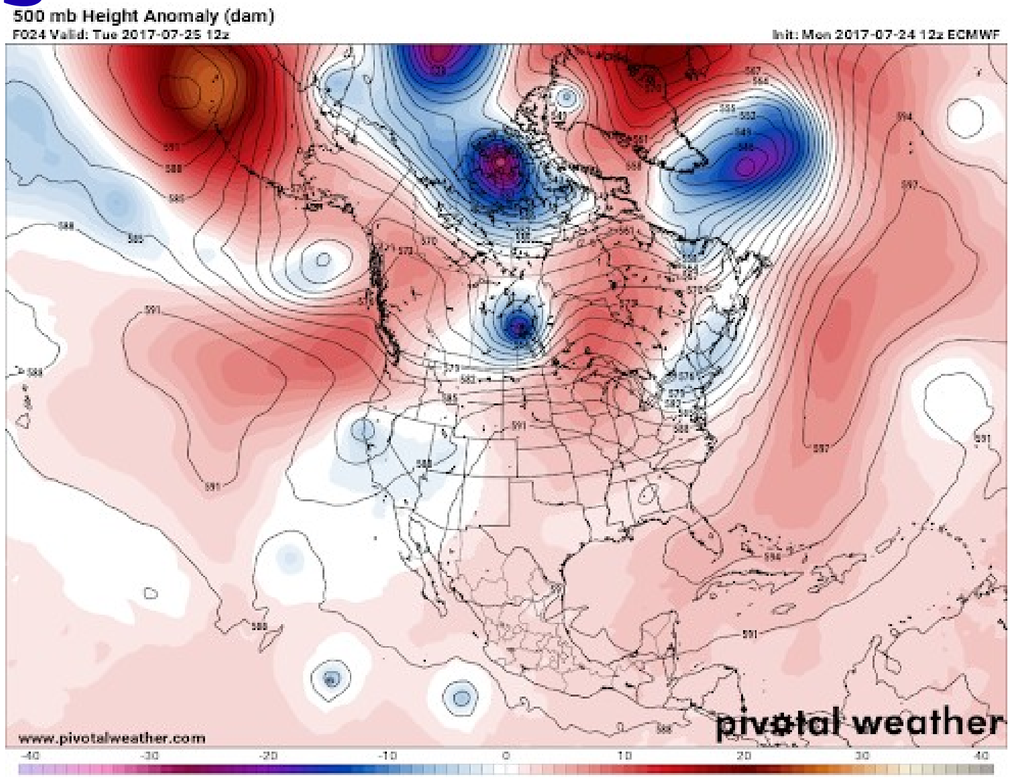
Input → historical Weather data
Process → Stability prediction model
Output → Pilot weather guidance document



Dependencies

- Forecast verification module
- Weather document generation syste

AI-BASED HYPER-LOCAL AVIATION WEATHER FORECAST VERIFICATION SYSTEM FOR SRI LANKAN MONSOON PATTERNS



Ability of Commercialization

- Can be used by airports and aviation authorities for weather monitoring.
- Supports safer flight operations by providing early weather alerts.
- Can be integrated with airport meteorological systems.

Material / Technical Cost

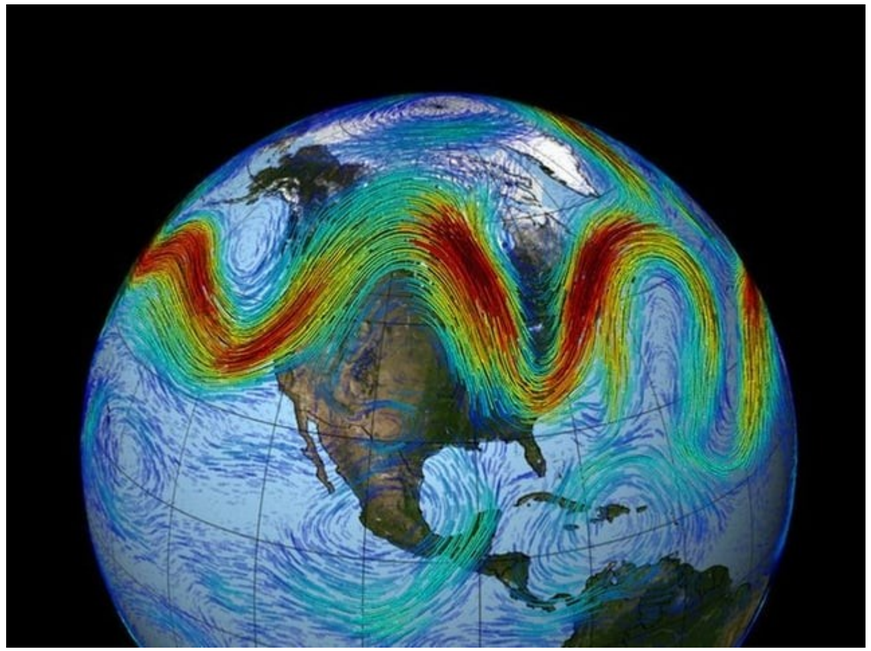
- **Cloud computing resources**
- **Data storage and system maintenance**

Labour Cost

- **AI model development**
- **Data analysis and system development**

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AI-BASED HYPER-LOCAL AVIATION WEATHER FORECAST VERIFICATION SYSTEM FOR SRI LANKAN MONSOON PATTERNS



Cost Recovery / Pricing

- Software subscription for airports and airlines
- Licensing for aviation authorities
- Integration service for aviation weather systems



Target Market

- **Airports in Sri Lanka**
- **Aviation meteorology departments**

AI-BASED HYPER-LOCAL AVIATION WEATHER FORECAST VERIFICATION SYSTEM FOR SRI LANKAN MONSOON PATTERNS



Value Proposition

- Early detection of hazardous weather conditions
- Improved aviation safety and operational planning
- More accurate airport-level weather analysis



Intellectual Property Rights

- **Machine learning model architecture**
- **Weather prediction algorithms**
- **Software system design**

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AI-BASED HYPER-LOCAL AVIATION WEATHER FORECAST VERIFICATION SYSTEM FOR SRI LANKAN MONSOON PATTERNS

GANTT CHART

Task	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Literature Review & Problem identification										
Proposal Preparation & Submission										
Data Collection (Airport Meteorological Data)										
Data Preprocessing & Feature Engineering										
Development of Coupled Prediction Model										
Model Evaluation & Forecast verification										
Weather Visualization Development										
System Testing & Result Validation										
Final Report & Website Development										
Final Presentation & Viva										
Research Paper Submission										
Research Paper Publication Evidence										

AI-BASED HYPER-LOCAL AVIATION WEATHER FORECAST VERIFICATION SYSTEM FOR SRI LANKAN MONSOON PATTERNS

Work Breakdown

Task ID	Task Description	Responsible Member	Timeline	Deliverable
1	Literature Review & Problem Identification	All Members	Mar – Apr	Literature review report, problem statement, objectives
2	Proposal Preparation & Submission	All Members	Apr	Approved research proposal
3	Data Collection (Airport Meteorological Data)	All Members	Apr – May	Historical weather data for NE, SW, First & Second Inter-Monsoons
4	Data Preprocessing & Feature Engineering	All Members	May – Jun	Cleaned and processed dataset, features extracted for ML (wind, pressure, humidity, temperature, cloud ceiling, visibility)
5	Development of Pressure-Humidity-Temperature Coupled Prediction Model	Sankadeepa W.D.I (IT22113986)	Jun – Aug	Multi-variable ML model for NE Monsoon, Weather Visualization (temperature heatmaps, wind pattern maps)
6	Development of Wind Pattern Prediction Model (SW Monsoon)	Ranasingha R.A.A.D (IT22269720)	Jun – Aug	Wind Pattern Prediction Model, Runway-Oriented Wind Prediction System, Pilot Portal for flight input and authentication
7	Development of Rapid Weather Change Detection (First Inter-Monsoon)	Jayawardhana H.M.S.I (IT22619976)	Jun – Aug	Rapid Weather Change Detection Model, Flight Time Processing Module, Cloud & Visibility Prediction Model, Threshold Alerts for "Below Minimum"
8	Development of Humidity & QNH Stability Model (Second Inter-Monsoon)	Jayathilaka P.K.V.K (IT22069504)	Jul – Sep	ML model predicting atmospheric stability based on relative humidity and pressure, reliability scoring
9	Weather Document Generation Module	Jayathilaka P.K.V.K (IT22069504)	Sep – Oct	Structured Pilot Guidance Document, Organized Weather Reports for pilots
10	Model Evaluation & Forecast Verification	All Members	Aug – Oct	Verification of accuracy for all models, forecast validation reports
11	Weather Visualization & Dashboard Development	All Members	Sep – Nov	Interactive visualizations for pilots (charts, alerts, dashboards)
12	System Integration, Testing & Result Validation	All Members	Oct – Nov	Fully integrated system, validated predictions and alerts
13	Final Report & Website Development	All Members	Nov – Dec	Final report, project website with dashboard and documentation
14	Final Presentation & Viva	All Members	Dec	Presentation slides, viva-ready system demonstration
15	Research Paper Submission	All Members	Dec	Submitted research paper to journal/conference
16	Research Paper Publication Evidence	All Members	Dec	Proof of publication or acceptance letter

AI-BASED HYPER-LOCAL AVIATION WEATHER FORECAST VERIFICATION SYSTEM FOR SRI LANKAN MONSOON PATTERNS

Collected Data set So far

Year	Month	Date	METAR	Time(U	Wind D	Wind sp	Visibilit	Weathe	Clouds	Dry ten	Dew po	RH(%)	QNH (h
2019	1	1	M	0010	030	07	9000		SKC	22.5	20.8	88	1015.2
2019	1	1	M	0040	030	07	9000		SKC	22.6	20.5	88	1015.5
2019	1	1	M	0110	030	08	8000		SKC	22.8	20.6	87	1015.6
2019	1	1	M	0140	030	09	8000		SKC	23.1	20.7	86	1015.9
2019	1	1	M	0210	030	11	9000		NSC	23.9	20.7	81	1016.3
2019	1	1	M	0240	040	11	9999		CAVOK	24.8	20.5	77	1016.6
2019	1	1	M	0310	040	11	9999		CAVOK	25.1	20.1	74	1016.7
2019	1	1	M	0340	040	14	9999		CAVOK	26.6	20.3	68	1016.6
2019	1	1	M	0410	040	12	9999		FEW018	27.0	20.7	68	1016.6
2019	1	1	M	0440	040	14	9999		FEW018	27.8	20.3	63	1016.5
2019	1	1	M	0510	030	12	9999		FEW018	28.3	20.0	60	1016.1
2019	1	1	M	0540	040	13	9999		FEW020	29.0	19.8	56	1015.7
2019	1	1	M	0610	040	12	9999		FEW020	30.1	20.0	54	1015.3
2019	1	1	M	0640	050	12	9999		FEW018	29.9	20.7	57	1014.7
2019	1	1	M	0710	030	12	9999		FEW018	29.9	19.8	54	1014.2
2019	1	1	M	0740	040	12	9999		FEW018	29.6	20.1	56	1013.3

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
106205	2024	12	31	M	1710	040	08	9999		FEW017 OVC100	25.1	24.4	96	1011.9
106206	2024	12	31	M	1740	VRB	02	9999		FEW017 OVC100	25	24.5	97	1011.7
106207	2024	12	31	M	1810	030	04	9999		FEW017 OVC100	24.8	24.3	97	1011.9
106208	2024	12	31	M	1840	030	03	9999		FEW017	24.7	24.2	97	1011.5
106209	2024	12	31	M	1910	030	04	9999		SCT017	24.7	24.4	98	1011
106210	2024	12	31	M	1940	050	05	9999		SCT017	24.7	24.4	98	1010.3
106211	2024	12	31	M	2010	110	02	9000		FEW017	24.5	24.3	99	1009.9
106212	2024	12	31	M	2040	140	04	9999		FEW017	24.3	24.1	99	1009.7
106213	2024	12	31	M	2110	VRB	02	9000		FEW017	24.1	24.1	100	1009.4
106214	2024	12	31	M	2140	170	03	9999		FEW017	24.2	24.2	100	1009
106215	2024	12	31	M	2210	080	03	9999		FEW017	24.2	24.2	100	1008.5
106216	2024	12	31	M	2240	010	04	9999		FEW017	24.1	24.1	100	1008.7
106217	2024	12	31	M	2310	020	05	9999		FEW017	24.3	24.1	99	1008.8
106218	2024	12	31	M	2340	020	05	9999		FEW017	24.5	24.3	99	1008.9
106219														
106220														
106221														
106222														
106223														

Thank You

